

K-디지털 트레이닝

비즈니스 계층

[KB] IT's Your Life

1 비즈니스 계층의 설정

✓ DTO, Data Transfer Object

○ VO, Value Object

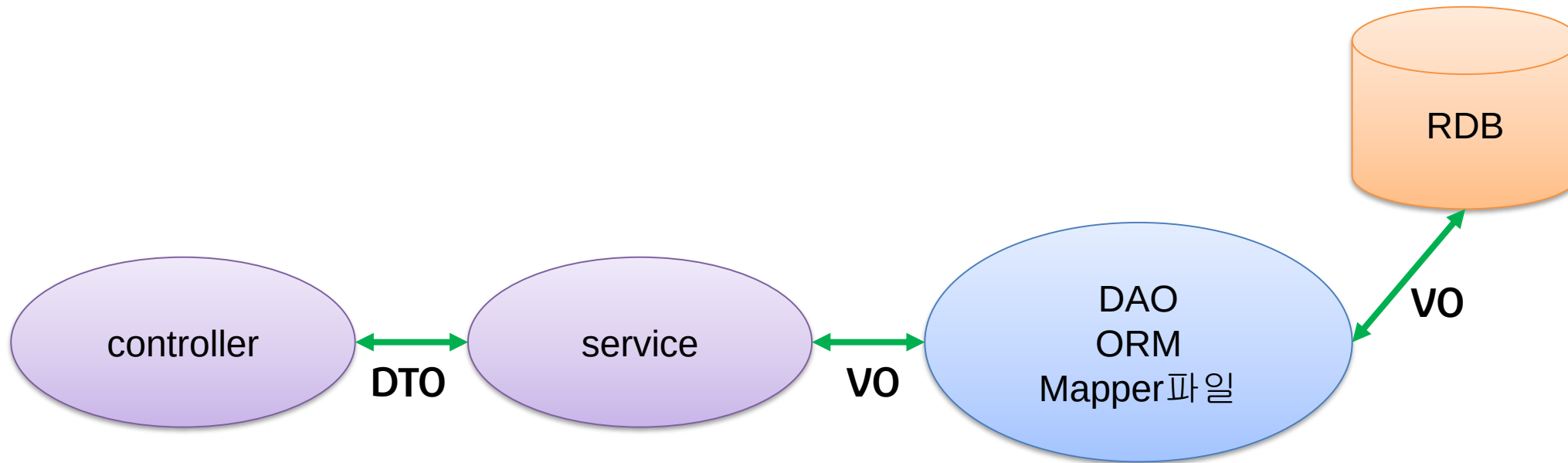
- 데이터베이스 테이블의 모양을 그대로 반영한 객체
- Dao에서 사용

○ DTO

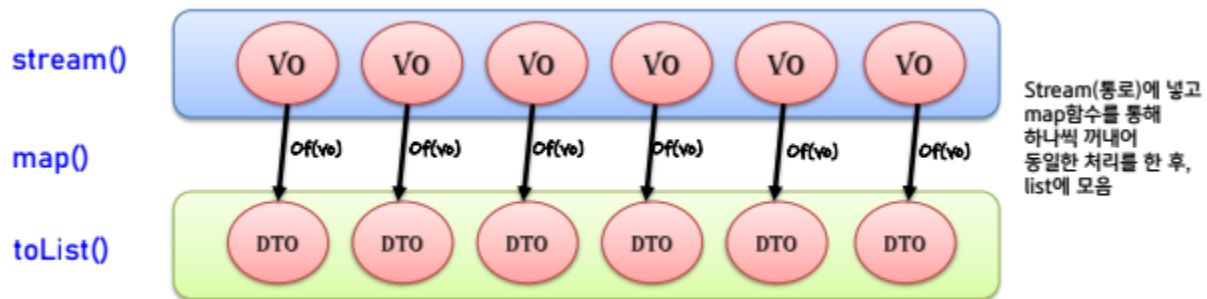
- Dao를 제외한 나머지 계층(Service, Controller, View 등)에서 사용
- 테이블에 없지만 비즈니스 로직에 필요한 추가 정보도 포함 가능

○ VO와 DTO간 상호 변환 기능 필요

- VO → DTO
- DTO → VO



변환할 것이 많은 경우 편하게 하는 방법



코드로 어떻게 만드나
(DTO class내에 함수 정의)

BoardDTO.java

```
// VO → DTO 변환
public static BoardDTO of(BoardVO vo) {
    return vo == null ? null : BoardDTO.builder()
        .no(vo.getNo())
        .title(vo.getTitle())
        .content(vo.getContent())
        .writer(vo.getWriter())
        .regDate(vo.getRegDate())
        .updateDate(vo.getUpdateDate())
        .build();
}

// DTO → VO 변환
public BoardVO toVo() {
    return BoardVO.builder()
        .no(no)
        .title(title)
        .content(content)
        .writer(writer)
        .regDate(regDate)
        .updateDate(updateDate)
        .build();
}
```



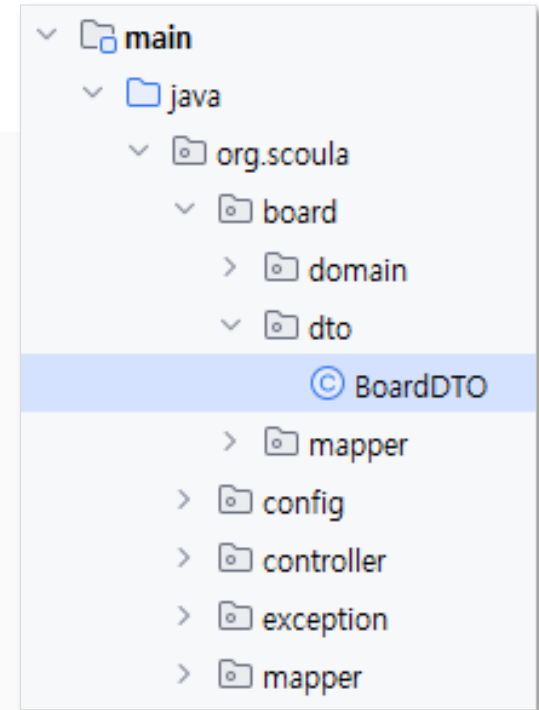
1 비즈니스 계층의 설정

BoardDTO.java

```
package org.scoula.board.dto;  
...
```

```
import java.util.ArrayList;  
import java.util.Date;  
import java.util.List;
```

```
@Data  
@NoArgsConstructor  
@AllArgsConstructor  
@Builder  
public class BoardDTO {  
    private Long no;  
    private String title;  
    private String content;  
    private String writer;  
    private Date regDate;  
    private Date updateDate;
```

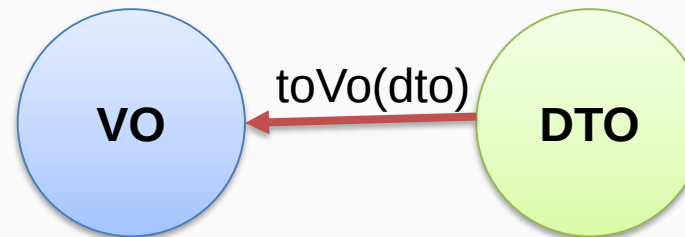
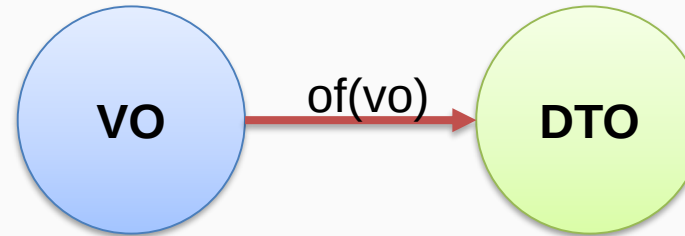


1 비즈니스 계층의 설정

BoardDTO.java

```
// VO → DTO 변환
public static BoardDTO of(BoardVO vo) {
    return vo == null ? null : BoardDTO.builder()
        .no(vo.getNo())
        .title(vo.getTitle())
        .content(vo.getContent())
        .writer(vo.getWriter())
        .regDate(vo.getRegDate())
        .updateDate(vo.getUpdateDate())
        .build();
}

// DTO → VO 변환
public BoardVO toVo() {
    return BoardVO.builder()
        .no(no)
        .title(title)
        .content(content)
        .writer(writer)
        .regDate(regDate)
        .updateDate(updateDate)
        .build();
}
}
```



1 비즈니스 계층의 설정

RootConfig.java

```
package org.scoula.config;

import javax.sql.DataSource;

import org.apache.ibatis.session.SqlSessionFactory;
import org.mybatis.spring.SqlSessionFactoryBean;
import org.mybatis.spring.annotation.MapperScan;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.Configuration;

import com.zaxxer.hikari.HikariConfig;
import com.zaxxer.hikari.HikariDataSource;

@Configuration
@PropertySource({"classpath:/application.properties"})
@MapperScan(basePackages = {"org.scoula.board.mapper"})
@ComponentScan(basePackages={ "org.scoula.board.service" })
public class RootConfig {

    ...

}
```

1 비즈니스 계층의 설정

BoardService.java

```
package org.scoula.board.service;

import org.scoula.board.dto.BoardDTO;

import java.util.List;
import java.util.Optional;

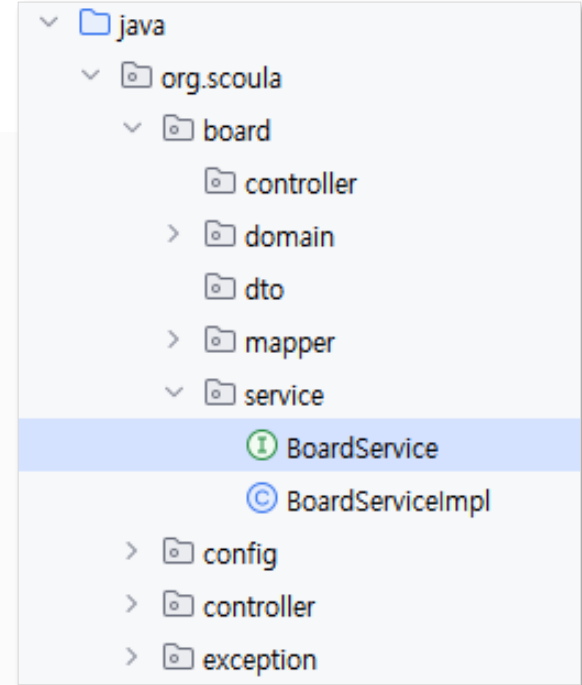
public interface BoardService {
    public List<BoardDTO> getList();

    public BoardDTO get(Long no);

    public void create(BoardDTO board);

    public boolean update(BoardDTO board);

    public boolean delete(Long no);
}
```



1 비즈니스 계층의 설정

 BoardServiceImpl.java

```
package org.scoula.board.service;
```

```
import java.util.List;
```

```
import org.springframework.stereotype.Service;
import org.scoula.board.domain.BoardVO;
import org.scoula.board.mapper.BoardMapper;
```

```
import lombok.AllArgsConstructor;
import lombok.extern.log4j.Log4j2;
```

```
@Log4j2
```

```
@Service
```

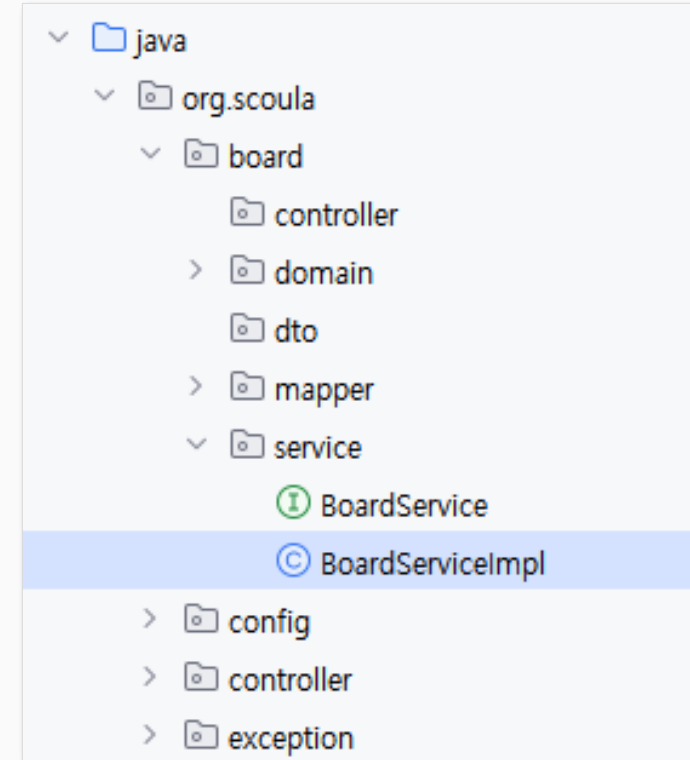
```
@RequiredArgsConstructor
```

```
public class BoardServiceImpl implements BoardService {
```

```
    final private BoardMapper mapper;
```

final 멤버를 인자로 가지는 생성자 추가

생성자가 1개인 경우
생성자 주입으로 초기화



1 비즈니스 계층의 설정

BoardServiceImpl.java

```
@Override
public List<BoardDTO> getList() {
    return List.of();
}

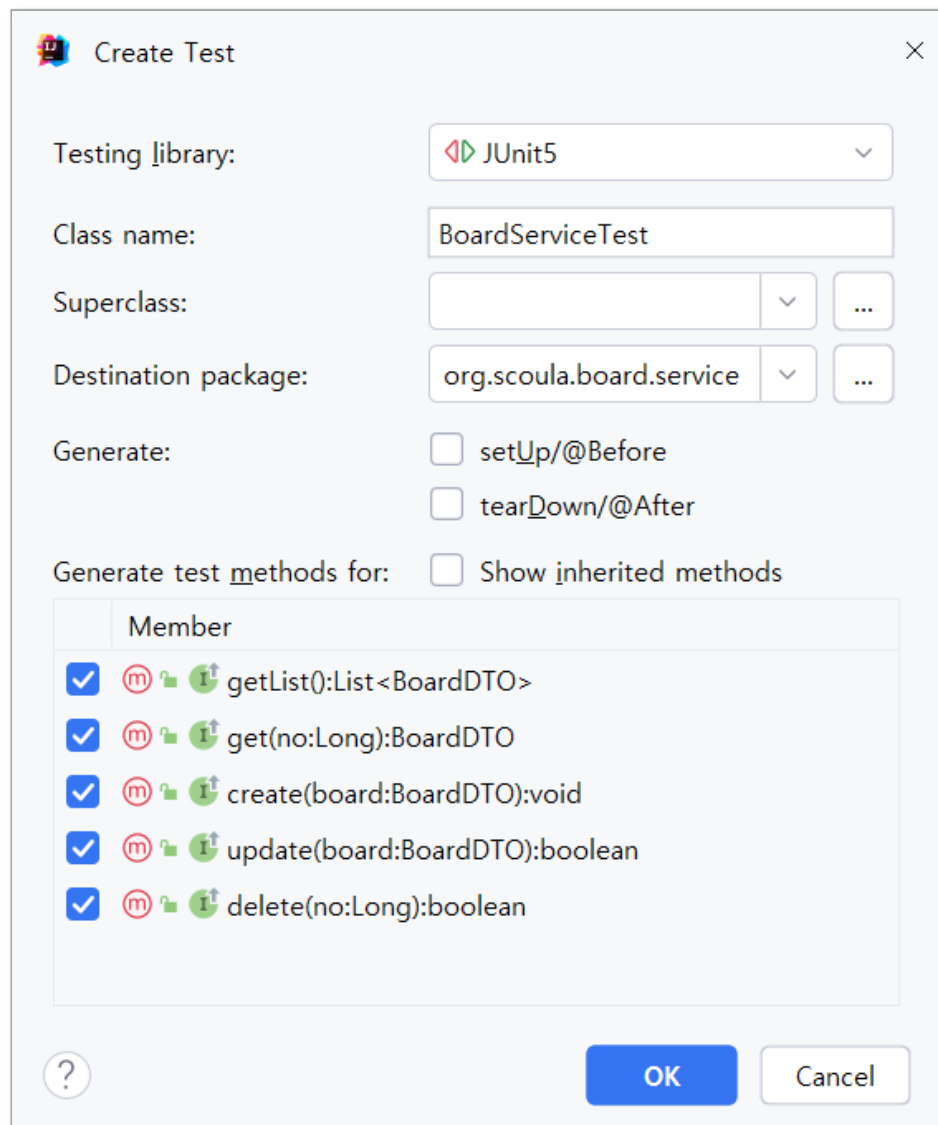
@Override
public BoardDTO get(Long no) {
    return null;
}

@Override
public void create(BoardDTO board) {
}

@Override
public boolean update(BoardDTO board) {
    return false;
}

@Override
public boolean delete(Long no) {
    return false;
}
}
```

✓ test :: BoardServiceTest.java



The image shows a 'Create Test' dialog box with the following fields and options:

- Testing library: JUnit5
- Class name: BoardServiceTest
- Superclass: (empty)
- Destination package: org.scoula.board.service
- Generate: ☐ setUp/@Before, ☐ tearDown/@After
- Generate test methods for: ☐ Show inherited methods

	Member
☒	☒ ☒ ☒ getList():List<BoardDTO>
☒	☒ ☒ ☒ get(no:Long):BoardDTO
☒	☒ ☒ ☒ create(board:BoardDTO):void
☒	☒ ☒ ☒ update(board:BoardDTO):boolean
☒	☒ ☒ ☒ delete(no:Long):boolean

Buttons: ? OK Cancel

test :: BoardServiceTest.java

...

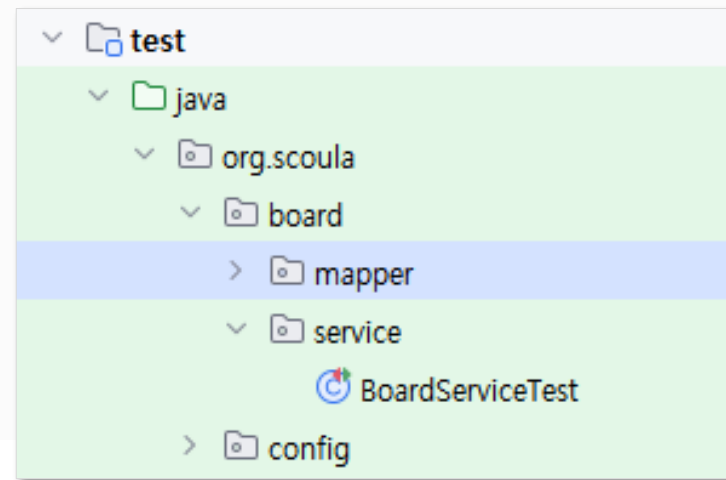
```
@ExtendWith(SpringExtension.class)
@ContextConfiguration(classes = {RootConfig.class} )
@Log4j2
```

```
public class BoardServiceTest {
```

```
    @Autowired
    private BoardService service;
```

```
    ...
```

```
}
```

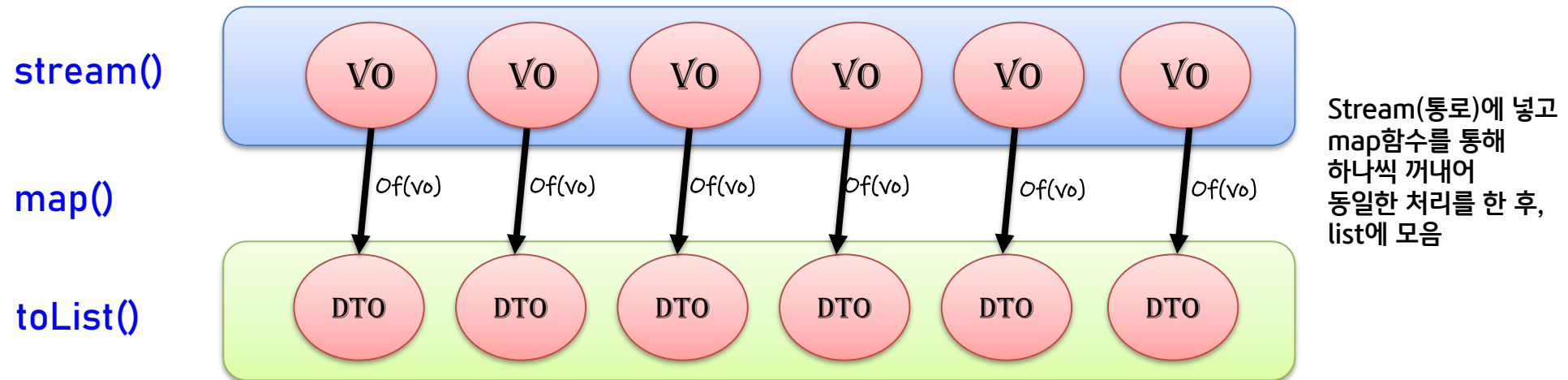


BoardServiceImpl.java

```
@Override
public List<BoardDTO> getList() {

    log.info("getList.....");

    return mapper.getList().stream() // BoardVO의 스트림
        .map(BoardDTO::of)           // BoardDTO의 스트림
        .toList();                   // List<BoardDTO> 변환
}
```



 test :: BoardServiceTest.java

```
@Test
public void getList() {

    for(BoardDTO board: service.getList()) {
        log.info(board);
    }

}
```

INFO org.scoula.board.service.BoardServiceImpl(getList:20) - getList.....
INFO jdbc.sqlonly(sqlOccurred:228) - **select * from tbl_board order by no desc**

INFO org.scoula.board.service.BoardServiceTest(getList:26) - BoardDTO(no=7, title=새로 작성하는 글, content=새로 작성하는 내용, writer=user0, regDate=Wed Jan 15 12:56:07 KST 2025, updateDate=Wed Jan 15 12:56:07 KST 2025)
INFO org.scoula.board.service.BoardServiceTest(getList:26) - BoardDTO(no=6, title=새로 작성하는 글, content=새로 작성하는 내용, writer=user0, regDate=Wed Jan 15 12:52:46 KST 2025, updateDate=Wed Jan 15 12:52:46 KST 2025)
INFO org.scoula.board.service.BoardServiceTest(getList:26) - BoardDTO(no=5, title=수정된 제목, content=수정된 내용, writer=user00, regDate=Wed Jan 15 11:53:10 KST 2025, updateDate=Wed Jan 15 12:57:42 KST 2025)
INFO org.scoula.board.service.BoardServiceTest(getList:26) - BoardDTO(no=4, title=테스트 제목4, content=테스트 내용4, writer=user00, regDate=Wed Jan 15 11:53:10 KST 2025, updateDate=Wed Jan 15 11:53:10 KST 2025)
INFO org.scoula.board.service.BoardServiceTest(getList:26) - BoardDTO(no=2, title=테스트 제목2, content=테스트 내용2, writer=user00, regDate=Wed Jan 15 11:53:10 KST 2025, updateDate=Wed Jan 15 11:53:10 KST 2025)
INFO org.scoula.board.service.BoardServiceTest(getList:26) - BoardDTO(no=1, title=테스트 제목1, content=테스트 내용1, writer=user00, regDate=Wed Jan 15 11:53:10 KST 2025, updateDate=Wed Jan 15 11:53:10 KST 2025)

BoardServiceImpl.java

```
@Override
public BoardDTO get(Long no) {

    log.info("get....." + no);

    BoardDTO board = BoardDTO.of( mapper.get(no));
    return Optional.ofNullable(board)
        .orElseThrow(NoSuchElementException::new);
}
```

 test :: BoardServiceTest.java

```
@Test
void get() {
    log.info(service.get(1L));
}
```

```
INFO org.scoula.board.service.BoardServiceImpl(get:31) - get.....1
INFO jdbc.sqlonly(sqlOccurred:228) - select * from tbl_board where no = 1
```

```
INFO org.scoula.board.service.BoardServiceTest(get:32) - BoardDTO(no=1, title=테스트 제목1, content=테스트 내용1, writer=user00, regDate=Wed Jan 15 11:53:10 K
ST 2025, updateDate=Wed Jan 15 11:53:10 KST 2025)
```


BoardServiceImpl.java

```
@Override
public void create(BoardDTO board) {

    log.info("create....." + board);

    BoardVO vo = board.toVo();
    mapper.create(vo);
    board.setNo(vo.getNo());
}
```

 test :: BoardServiceTest.java

```
@Test
public void create() {

    BoardDTO board = new BoardDTO();
    board.setTitle("새로 작성하는 글");
    board.setContent("새로 작성하는 내용");
    board.setWriter("user1");

    service.create(board);

    log.info("생성된 게시물의 번호: " + board.getNo());
}
```

INFO org.scoula.board.service.BoardServiceImpl(create:43)

- create.....BoardDTO(no=null, title=새로 작성하는 글, content=새로 작성하는 내용, writer=user1, regDate=null, updateDate=null)

INFO jdbc.sqlonly(sqlOccurred:228) - **insert into tbl_board (title, content, writer) values ('새로 작성하는 글', '새로 작성하는 내용', 'user1')**

INFO jdbc.sqlonly(sqlOccurred:228) - **SELECT LAST_INSERT_ID()**

INFO org.scoula.board.service.BoardServiceTest(create:45) - 생성된 게시물의 번호: 8

BoardServiceImpl.java

```
@Override
public boolean update(BoardDTO board) {

    log.info("update....." + board);

    return mapper.update(board.toVo()) == 1;
}
```

 test :: BoardServiceTest.java

```
@Test
public void update() {

    BoardDTO board = service.get(1L);

    board.setTitle("제목 수정합니다.");
    log.info("update RESULT: " + service.update(board));
}
```

INFO org.scoula.board.service.BoardServiceImpl(get:32) - get.....1

INFO jdbc.sqlonly(sqlOccurred:228) - select * from tbl_board where no = 1

INFO org.scoula.board.service.BoardServiceImpl(update:52) - update.....BoardDTO(no=1, title=제목 수정합니다., content=테스트 내용1, writer=user00, regDate=Wed Jan 15 11:53:10 KST 2025, updateDate=Wed Jan 15 11:53:10 KST 2025)

INFO jdbc.sqlonly(sqlOccurred:228) - **update tbl_board set title = '제목 수정합니다.', content = '테스트 내용1', writer = 'user00', update_date = now() where no = 1**

INFO org.scoula.board.service.BoardServiceTest(update:54) - update RESULT: true

BoardServiceImpl.java

```
@Override
public boolean delete(Long no) {

    log.info("delete...." + no);

    return mapper.delete(no) == 1;
}
```

 test :: BoardServiceTest.java

```
@Test
public void delete() {

    // 게시물 번호의 존재 여부를 확인하고 테스트할 것
    log.info("delete RESULT: " + service.delete(2L));

}
```

INFO org.scoula.board.service.BoardServiceImpl(delete:60) - delete....2

INFO jdbc.sqlonly(sqlOccurred:228) - **delete from tbl_board where no = 2**

INFO org.scoula.board.service.BoardServiceTest(delete:61) - delete RESULT: true